

Customer Shopping Behaviour Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 customer purchases across multiple product categories. The objective is to identify spending patterns, customer preferences, subscription trends, and revenue-driving factors through data analytics. Python was used for data preparation, PostgreSQL for data analysis, and Power BI for visualization. The insights generated help businesses understand customer behavior and make informed decisions to improve sales and customer engagement.

2. Dataset Summary

The dataset used in this project contains **3,900 customer purchase records and 18 features (columns)** related to retail shopping behavior. It includes customer demographics (Age, Gender), product information (Item Purchased, Category, Size, Color, Season), transaction details (Purchase Amount, Payment Method, Shipping Type), and customer engagement metrics (Review Rating, Subscription Status, Discount Applied, Promo Code Used, and Previous Purchases). The dataset provides a comprehensive view of customer purchasing patterns and enables the analysis of spending behavior, product preferences, and business performance.

3. Exploratory Data Analysis using Python

Data preparation and exploratory analysis were conducted using Python to understand the dataset structure, improve data quality, and prepare the data for database integration and visualization.

- **Data Loading:** The customer shopping behavior dataset was imported using the Pandas library.
- **Initial Exploration:** Dataset structure, data types, and record counts were examined using `df.info()`, while descriptive statistics were generated using `b`.
- **Missing Data Handling:** Missing values were identified and analyzed. The Review Rating column was treated to ensure data completeness and consistency.
- **Column Standardization:** Column names were converted to a standardized snake_case format to improve readability, maintain consistency, and support database operations.
- **Feature Engineering:**
 - An age_group feature was created by categorizing customers into different age segments.

- A `purchase_frequency_days` feature was generated by converting purchase frequency categories into numerical day values.
- **Data Consistency Check:** The relationship between `discount_applied` and `promo_code_used` was evaluated, and the redundant `promo_code_used` column was removed to eliminate duplicate information.
- **Database Integration:** The cleaned and transformed dataset was exported to a PostgreSQL database to support advanced SQL analysis and business intelligence reporting.

4. Data Analysis Using SQL

SQL queries were executed on the PostgreSQL database to analyze customer behavior, purchasing patterns, and business performance metrics.

I. **Revenue by Gender** – Compared total revenue generated by male and female customers.

II. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839 Query complete 00:00:00.271		

	gender text	revenue numeric
1	Female	75191
2	Male	157890

III. **Top-Rated Products** – Determine the top 5 products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

IV. Purchase Amount by Shipping Type – Compared average purchase amounts between Standard and Express shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

V. Subscriber vs. Non-Subscriber Spending – Analyzed average spending and total revenue generated by subscribed and non-subscribed customers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

VI. Products with Highest Discount Usage – Identified the top 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

VII. Customer Segmentation Analysis – Categorized customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

VIII. Top Products by Category – Identified the top 3 most purchased products within each product category.

	item_rank bigint	category text	item_purchased text	totalOrders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
Total rows: 11		Query complete 00:00:00.127		

IX. Repeat Buyer Subscription Analysis – Examined whether repeat buyers were more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

X. Revenue Contribution by Age Group – Evaluated revenue generated by different age groups to identify high-value customer segments.

	age_group text	total_revenue numeric
1	Senior	88480
2	Middle Age	65629
3	Adult	44342
4	Young	34630

5. Dashboard in Power BI

An interactive Power BI dashboard was created to visualize customer shopping behavior and key business insights. The dashboard includes KPI cards, category-wise revenue and sales analysis, age group analysis, subscription status distribution, and interactive filters. It enables users to explore customer trends and supports data-driven decision-making through intuitive visualizations.





6. Business Recommendations

- Increase Subscription Adoption – Offer exclusive benefits to attract more subscribers.

- Focus on Top Categories – Invest more in high-performing product categories.
- Target High-Value Customers – Create personalized campaigns for major revenue-contributing age groups.
- Improve Customer Retention – Introduce loyalty programs to encourage repeat purchases.
- Optimize Discounts – Use targeted discounts to boost sales without reducing profitability.
- Enhance Customer Experience – Improve products and services based on customer reviews.
- Leverage Data Analytics – Use customer insights to support marketing and business decisions.